

Lava from a volcano can reach 2,282°F (1,250°C)—**12 times** hotter than boiling water.

INTO THE VOLCANO

FORECASTING VOLCANIC ERUPTIONS

When volcanoes erupt, they spit out dangerous hot ash, lava, and poisonous gases. In 2019, more than 29 million people were living near active volcanoes, so predicting eruptions is important. While it is difficult to know exactly when a volcano will erupt, vulcanologists gather clues by measuring earthquakes in volcanic regions, using satellites in space to monitor volcanoes and their surroundings, and even climbing down into them to collect gas samples.



VOLCANIC ACTIVITY

Volcanoes can either be extinct, meaning unlikely to erupt again; active, which means they erupt frequently; or dormant, meaning they have not erupted for some time. The Chaitén Volcano in Chile hadn't erupted for more than 350 years before erupting in 2008.

HOW VOLCANOES FORM

A volcano is formed where molten rock breaks through Earth's crust. Below ground, molten rock is called magma, and above ground it is called lava. Erupting magma forms ash and a rock called pumice. Layers of cooled lava, pumice, and ash can form a volcanic mountain.

The magma chamber below the volcano contains molten rock.

Clouds of ash and dust form during a volcanic eruption.

Hot lava can flow, or pumice and ash can explode, from the crater.

Lava flows down slope.

Layers of cooled lava, pumice, and ash.

